DE LA SALLE UNIVERSITY-DASMARIÑAS
COLLEGE OF ENGINEERING, ARCHITECTURE, AND TECHNOLOGY
ENGINEERING DEPARTMENT
COMPUTER ENGINEERING PROGRAM

SOFTWARE DESIGN

(T-CPET3228LA)

Enabling Assessment 2 – Log-in and Dashboard

Grade

Submitted By:
Gutierrez, Corbin Ezekiel
Valencia, Aaron Daniel

Submitted To:

Engr. Michael C. Olivo
Professor

ASSESSMENT TASK:

1. login and dashboard with database (Role Based Access Control).

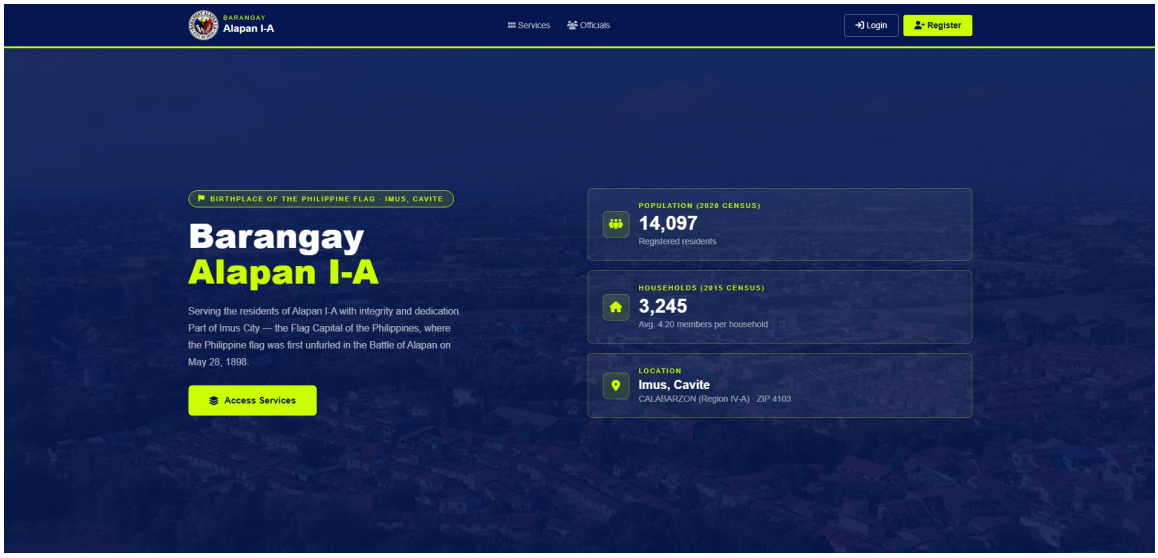
2. Chapter 1

A. Give all the screenshots of your dashboard and login

B. Explains each screenshot

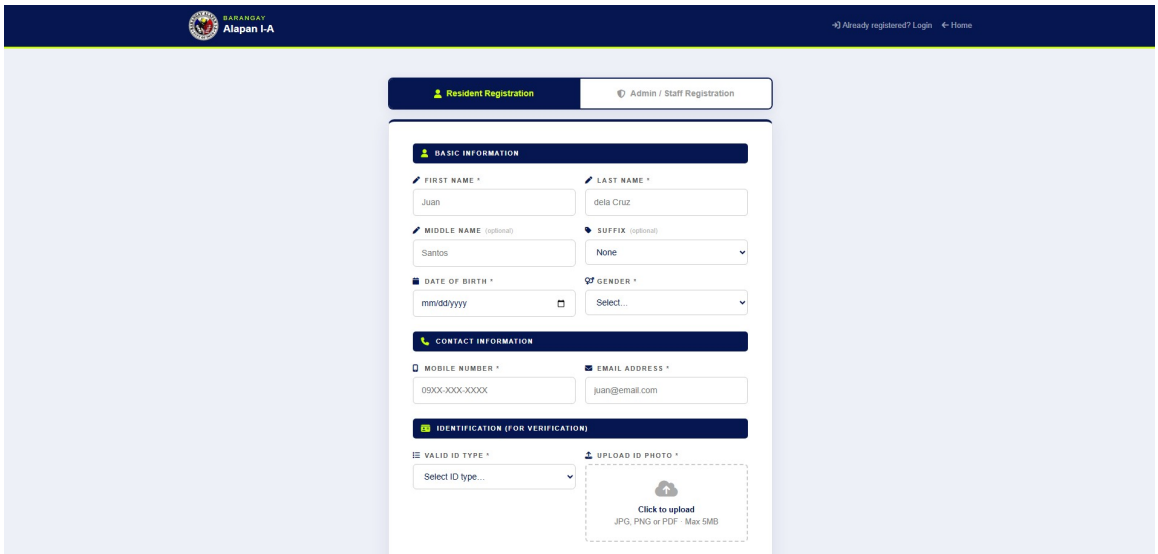
Rubrics

Criteria	Functionality and Completeness
Dashboard	40%
Login	40
Document	20%
Total	100%



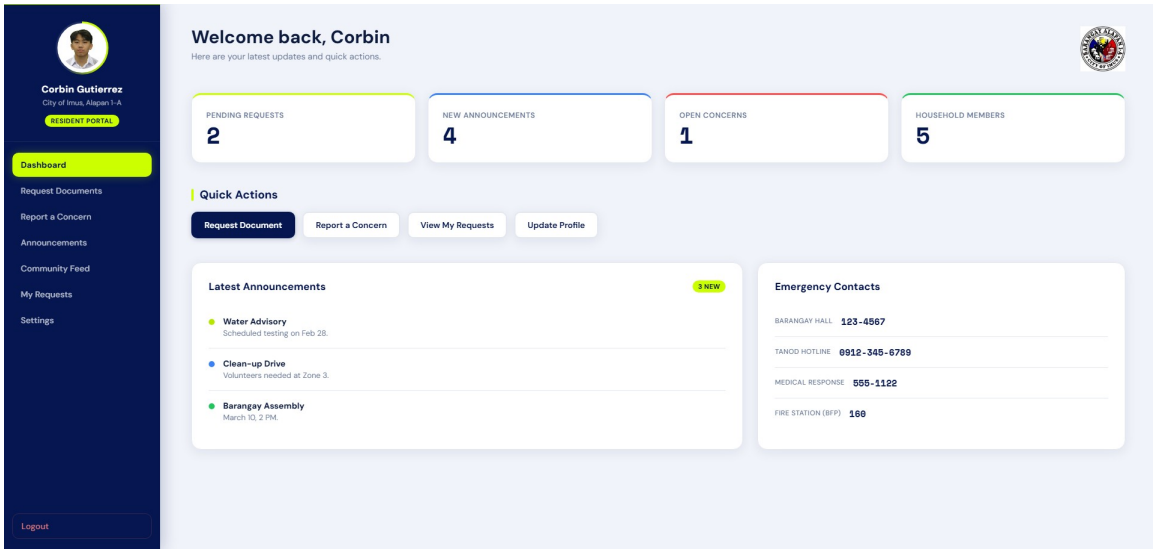
Home Page

It introduces the barangay to visitors with a hero section, key statistics, available services, and the list of officials. Anyone can view it, but clicking any service prompts a login gate, encouraging residents to register or log in before they can transact online.



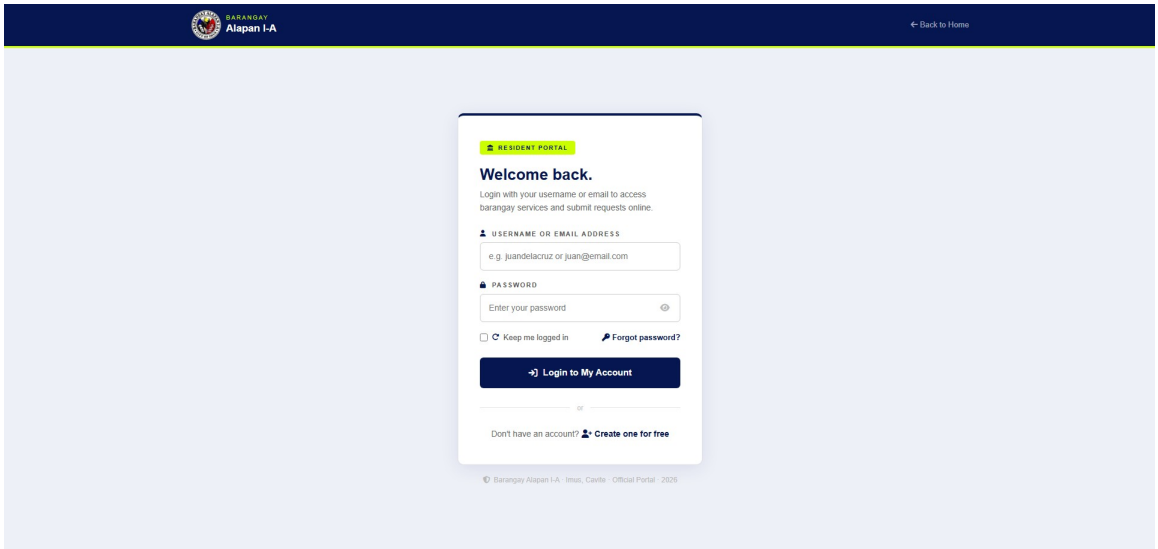
Registration Page

Where new residents and barangay staff create their accounts. Residents fill in basic personal info, contact details, and upload a valid ID for identity verification. Staff and admin use the same form but switch to the Admin/Staff tab, which adds a position field and a note to the Punong Barangay, since staff accounts require approval before access is granted.



Dashboard Page

It gives a quick overview of their pending document requests, announcements, open concerns, and household info. Quick action buttons let them request documents, report concerns, or update their profile without navigating away. It's the central hub for everything a logged-in resident needs to do.



Login Page

Accepts either a username or email alongside a password. A single login form serves all roles, residents, staff, and admins, and the system routes them to the appropriate dashboard based on their account type after authentication.

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Technology communication has reshaped how individuals and communities engage, share information, and participate in governance processes. Online platforms and e-government services are widely used to support citizen engagement, improve transparency, and provide efficient access to public services (Bangero, 2025). In the Philippines, barangays the smallest political units of local governance. However, many barangay operations remain heavily reliant on traditional or manual systems, which often hinder timely communication and active participation from residents (Alvero et al., 2025).

Several studies highlighted the potential benefits of digital systems in improving barangay services. For example, digital information systems have been shown to streamline administrative functions such as resident profiling, record-keeping, and certificate issuance.. Digitizing governance processes also enhances efficiency, transparency, and service delivery, contributing to stronger local governance structures (Aliling et al., 2025).

A descriptive study on ICT use in a Philippine barangay found that residents often use gadgets and digital devices, yet are less updated on local events due to limited information flow (Alvero et al., 2025). Yet current digital government practices often rely on generic platforms such as social media which are not officially monitored or tailored to barangay needs leading to fragmented communication (Bangero, 2025).

A digital public community space where residents can share ideas and engage in dialogue is one of the ideal e-governance system. By designing an integrated platform that allows not only access to e-services but also community interaction, barangay governance can transform from one-way announcements to participatory communication and collaboration.

BarangayKonek addresses these gaps by providing a barangay-centered social networking platform that integrates service access, resident interaction, and official communication in a single digital environment. This will enable residents to post updates, engage in discussions, and receive official announcements fostering stronger community engagement and collaboration.

1.2 Statement of the Problem

Despite the growing reliance on digital technologies, local communication and resident participation in barangay governance remain limited due to:

Absence of an official community interaction space. Residents currently depend on unofficial external social platforms that are not managed by barangay officials and lack contextual relevance to barangay affairs (Bangero, 2025).

Fragmented communication processes. Important announcements and information reach residents inconsistently due to lack of a centralized digital communication channel (Alvero et al., 2025).

Limited resident engagement. ICT use by residents does not fully translate to participation in barangay governance due to inadequate platforms catering to local dialogue and feedback (Alvero et al., 2025).

1.3 Objectives of the Study

General Objective

- To design and develop a barangay-centered social networking platform that allow community engagement through interactive communication and integrated digital services.

Specific Objectives

1. To develop a community feed for residents to post, comment, and interact.
2. To develop verified announcements and information system.
3. To create secure resident profiles with barangay verification.
4. To integrate barangay e-services (e.g., document requests, concern reporting).
5. To include moderation tools for content regulati
6. To evaluate the system's usability, accessibility, and impact on engagement.

1.4 Scope and Limitations

The study includes development of:

1. Resident verification and profile management system.
2. Community feed with posting, commenting, and reactions.
3. Official announcement board.
4. Document request and concern reporting modules.
5. Dashboards for residents, staff, and administrators.
6. Community tab for residents engagement

Limitations

1. Implementation limited to one barangay.
2. Focus on web-based platform; mobile app extension is excluded.
3. Private messaging between residents is optional or limited due to scope constraints.

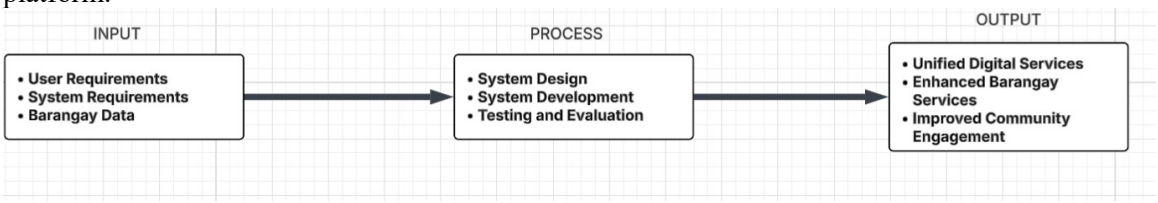
1.5 Significance of the Study

This study on the development of Barangay-Centered Social Networking Platform for Community Engagement. is expected to be significant and beneficial to the following:

Residents. Provides an officially moderated online space to interact, access services, and participate in local discussions.
Barangay Officials. Improves communication efficiency, transparency, and responsiveness to community needs.
Local Government Units. Offers a replicable model for digital governance that strengthens citizen engagement.
Researchers and Developers. Contributes to the body of knowledge on integrating social networking with e-government services.

1.6 Conceptual Framework

The conceptual framework of this study is based on the idea that community engagement increases when residents are provided with accessible communication channels, reliable information sources, and integrated digital services. BarangayKonek is designed to combine these three elements into one unified platform.



1.7 Definition of Terms

Community Engagement — Active participation of residents in communication and governance processes facilitated by digital platforms.
E-Governance — Use of ICTs to improve government operations and public services.
Resident Profile — Verified account associated with a barangay household.
Social Networking Platform — A digital medium that enables user interaction, content sharing, and community building.

References:

Aliling, R. J., Paclar, M. A., Abiera, J. P., & Solo, D. R. (2025). *Digitalizing governance: A transformation on the processes of one community in the Philippines*. ResearchGate. https://www.researchgate.net/publication/393348263

Alvero, A. C., Arizobal, J. A., Diano, R. C., Hernandez, J. S., Ladera, M. P., & Sim, C. J. (2025). *To explore the use and effects of ICT on barangay management and communication in Barangay Marawis, Iligan City*. E-Journals Philippines. https://ejournals.ph/article.php?id=25481

Bangero, J. C. (2025). *E-government practices and the role of digital platforms in governance*. ScienceDirect. https://www.sciencedirect.com/science/article/pii/S0740624X25000048

Digital Barangay Ayos Lomboy. (2025). *Web-based management and information system*. Scribd. https://www.scribd.com/document/868249583/Digital-Barangay-Ayos-Lomboy-s-Web-based-Management-and-Information-System

Digital Public Square. (n.d.). In *Wikipedia*. Retrieved 2026, from https://en.wikipedia.org/wiki/Digital_public_square